



The metamaterial sensors for multi-parameter measurement in structural health monitoring

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ABSTRACT

A multi-parameter metamaterial sensor is proposed to detect the positional variations of a metal wire, including rotational angle, vertical displacement, and horizontal displacement. The sensor consists of three layers: a metal ground, a dielectric layer (Arlon 880), and a metal structure. The metal structure includes a double resonant ring (DRR) and a microstrip line (ML). An equivalent circuit model and surface current distributions are used to analyze the sensor's physical mechanism. The results indicate that frequency shifts are caused by the coupling between the DRR and the metal wire (MW) at varying displacements, which facilitate the detection of rotational angle, vertical displacement, and horizontal displacement. The simulation results are consistent with the measurement data, confirming the sensor's high sensitivity to different MW displacements. The rotational angle sensitivity (0° – 180°) of the sensor is 4.06 MHz/degree, while its vertical displacement sensitivity (0 mm to 21 mm) is 18.9 MHz/mm. The horizontal displacement sensitivity varies in two ranges: 0–14 mm (16.4 MHz/mm) and 15–21 mm (29.6 MHz/mm). These findings demonstrate the sensor's potential for structural health monitoring applications.

1. Introduction

Metamaterials, as a class of artificial materials, are primarily composed of subwavelength structures that are artificially manipulated and arranged in specific configurations. These materials can exhibit many properties that are not found in natural materials, such as negative refractive index and superlens effects. Consequently, they can be used to design high-performance devices such as absorbers [1,2], metamaterial antennas [3], polarization converters [4], and filters [5]. Furthermore, metamaterials display evanescent wave properties [6,7], and their high sensitivity and resolution make them highly useful in the field of sensors [8–13].

With the continuous advancement of technology and society, industries such as aerospace, radar monitoring, shipbuilding, and automobile manufacturing are rapidly evolving. Metallic materials are widely used in these industries. However, prolonged use or environmental factors can cause small cracks or damage in these materials, compromising their structural integrity and safety. Structural health monitoring (SHM) provides an effective solution to ensure the safety of metallic materials by actively monitoring and measuring their

deformation [14].

Sensors play a critical role in the SHM systems by collecting and transmitting relevant monitoring signals [15]. These include optical fiber sensors [16], piezoelectric sensors [17], stress sensors [18], and temperature sensors [19]. Compared to these conventional sensors, metamaterial sensors [20–26] offer significant advantages such as high sensitivity, miniaturization, and strong anti-interference capabilities [20]. For example, V. G. M. Annamdas *et al.* [21] designed a novel robust metamaterial waveguide coupling zone for monitoring the horizontal and vertical in displacement monitoring of civil engineering prototype structures such as pipelines. This contact surface wave-based technique has been demonstrated to be at least ten times more sensitive than the electromechanical piezo-impedance-based SHM technique [22]. Similarly, P. Song *et al.* [23] proposed a nanocomposite piezoresistive sensor made of carbon black and polyvinyl pyrrolidone for monitoring loosening in composite bolted joints. C. Huang *et al.* [24] introduced an electromagnetically induced transparency (EIT)-like effect to achieve simultaneous communication and sensing using passive wireless antenna sensors. This sensor is used to monitor the growth of cracks. B. Ozbey *et al.* [25] designed a metamaterial sensor capable of detecting

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the displacement and deformation of reinforcing bars within a simply supported beam.

Despite these advancements, many existing sensors are limited to detecting a single parameter, which restricts their application scenarios. In the case of multi-parameter detection, the materials and structures of many resonator sensors cannot effectively respond to various types of physical or chemical stimuli simultaneously. This limitation hinders their ability to detect multiple parameters because some materials may react too strongly to certain parameters while showing insufficient response to others. By contrast, the ability to monitor multiple parameters can significantly broaden their range of applications and reduce monitoring costs. Therefore, designing a metamaterial sensor to detect multiple parameters simultaneously holds considerable potential for SHM applications.

This study presents a metamaterial sensor for multi-parameter measurement based on a microstrip line (ML)-loaded double resonant ring (DRR). The feasibility and effectiveness of the proposed sensor in the detection of rotation angle (RA), vertical displacement (VD), and horizontal displacement (HD) are validated through simulations and experimental measurements. By leveraging the sensor's ability to detect all three parameters, it can be applied in various scenarios, significantly reducing the monitoring costs. For example, the rotating parts of various industrial machines, such as generators, robotic arms, and Computer Numerical Control (CNC) lathes, require continuous monitoring to ensure optimal performance and operational safety. This continuous monitoring helps to promptly identify imbalances, wear, or other potential issues that could lead to machine failure. Additionally, monitoring the horizontal displacement of railway tracks is essential for maintaining proper track alignment. Ensuring the correct positioning of the tracks helps prevent accidents such as train derailments, which can occur due to misalignment or displacement. The aforementioned two examples involve the detection of rotational angles and horizontal displacements, respectively. Therefore, the proposed sensor is suitable for both applications. Its multi-functional capability broadens its application scope and significantly reduces the detection costs. Thus, the sensor exhibits immense potential for SHM applications.

2. Design and analysis

2.1. Multi-parameter measurement sensor design

Fig. 1(a) and 1(b) show the top view and side view of the sensor. As depicted in these two figures, the sensor consists of a metal ground (MD), a dielectric layer made of Arlon 880, and a metal structure that includes a DRR and a ML. The middle part of the ML exhibits a semi-circular shape, optimizing the coupling between the ML and the DRR.

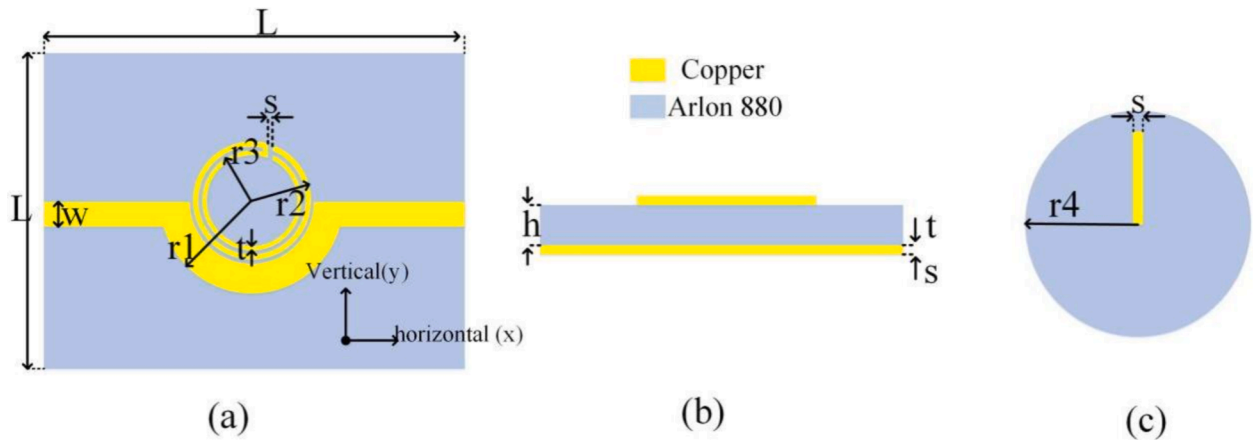


Fig. 1. Schematic of multi-parameter measurement sensor diagram (a) top view of the sensor, (b) side view of the sensor, and (c) a movable slab composed of a metal wire.

The enhanced coupling causes the DRR to respond more strongly to external stimuli. Fig. 1(c) illustrates a movable slab composed of a metal wire (MW). To facilitate the detection of the MW's displacement, the MW is etched onto the dielectric slab. Additionally, the dielectric slab has a minimal impact on the experimental results. The metal structure of the sensor is made of copper with a conductivity of 5.96×10^7 S/m. The Arlon 880 substrate has a relative dielectric constant of 2.2 with a loss tangent of 0.0009. The parameters shown in Fig. 1(a-c) are as follows: length of dielectric slab: $L = 90$ mm, thickness of metal structure: $t_s = 0.018$ mm, thickness of dielectric slab: $h = 2$ mm, outer radius of the ML's semicircular ring: $r_1 = 22.5$ mm, outer radius of the DRR's outer ring: $r_2 = 14.8$ mm, outer radius of the DRR's inner ring: $r_3 = 12.6$ mm, radius of the movable circular slab: $r_4 = 17.0$ mm, DRR width: $t = 0.9$ mm, width of ML: $w = 7.5$ mm; width of MW: $s = 1.0$ mm, length of MW: $l = 14.8$ mm.

2.2. Fundamental principles

The resonant characteristics of the sensor are analyzed using the principle of equivalent circuit. To illustrate the interaction between the sensor's metal structures, Fig. 2(a) presents the equivalent parameters of the DRR and ML [27,28]. In this model, L_1 and L_2 represent the equivalent inductance of the outer and inner rings of the DRR, respectively. L_3 is the equivalent inductance of the connecting line between the two rings of the DRR. L_4 represents the equivalent inductance of the ML. C_1 and C_2 represent the equivalent capacitances at the splits of the outer and inner rings of the DRR. C_3 denotes the equivalent capacitance between the DRR and the ML. Thus, the corresponding equivalent circuit model of the sensor is shown in Fig. 2(b). It is important to note that C_4 denotes the equivalent capacitance between the DRR and the MD, while C_5 represents the equivalent capacitance between the ML and the MD.

To verify the sensor model, electromagnetic simulations are performed using CST simulation software. The x, y, and z directions of the sensor are set as open boundaries, and planar waves are applied as the excitation source at both ends of the ML. The frequency range is set to 0.0–1.6 GHz. The equivalent circuit model in Fig. 2(b) is simulated and curve-fitted using the Advanced Design System (ADS) software. The parameter values for the equivalent circuit model are obtained by fitting the simulation curves. The simulation and calculated fitting results within the frequency range of 0.0 to 1.6 GHz are shown in Fig. 3. Both curves exhibit two resonance dips in this range. The resonant frequency of the low-frequency dip is 0.50 GHz, while that of the high-frequency dip is 1.18 GHz. Therefore, the results obtained using both electromagnetic simulation and the equivalent circuit model are consistent.

The low-frequency dip in the sensor's transmission curve is more sensitive to changes in the three parameters (RA, HD, and VD). To

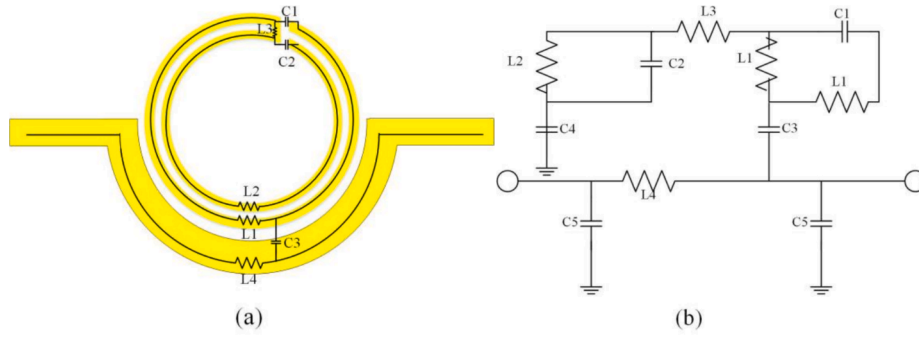


Fig. 2. Equivalent circuit diagram (a) simple schematic diagram depicting each parameter; (b) equivalent circuit model.

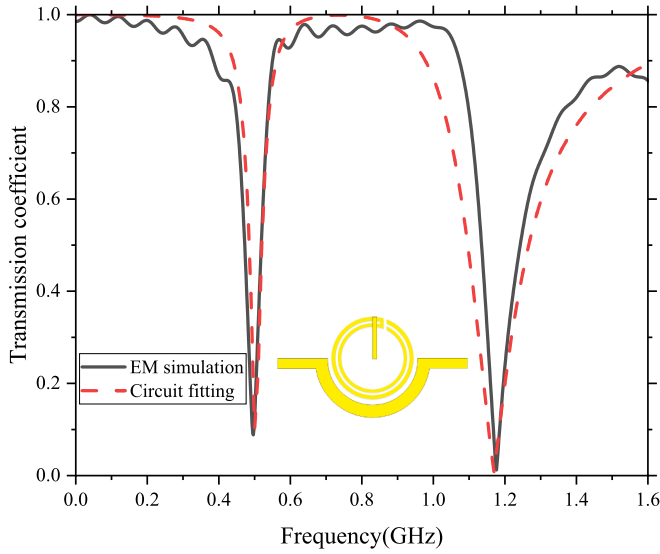


Fig. 3. Comparison of CST simulation and equivalent circuit simulation results. The circuit parameters are as follows: $L1 = 11.3\text{nH}$, $L2 = 1.8\text{nH}$, $L3 = 0.5\text{nH}$, $L4 = 3.7\text{nH}$, $C1 = 2.1\text{pF}$, $C2 = 4.8\text{pF}$, $C3 = 4.2\text{pF}$, $C4 = 10.9\text{pF}$, $C5 = 0.1\text{pF}$.

explain this, Fig. 4(a) and 4(b) show the simulated surface current distributions of the sensor at 0.50 GHz and 1.18 GHz, respectively, at the initial position. It is evident that the resonant frequency of the low-frequency dip is influenced by both the outer and inner rings of the DRR. By contrast, the resonant frequency of the high-frequency dip is primarily influenced by the outer ring, with minimal impact from the inner ring. Both the inner and outer rings contribute to the equivalent inductance of the DRR. Therefore, during the displacement of the MW, the change in the equivalent inductance associated with the low-frequency dip is greater than that for the high-frequency dip. This results in a higher sensitivity of the low-frequency dip to displacement changes. Consequently, only the variation in the resonant frequency of

the low-frequency dip is observed during the sensing process.

3. Simulation and experimental verification

The experiment setup of the sensor is shown in Fig. 5(a) and 5(b). The sensor is fabricated using printed circuit board (PCB) technology, with a layer of gold deposited on the copper surface through an electroless gold plating process. This gold layer acts as a protective coating to prevent oxidation. This method significantly delays the contact between copper and oxygen or moisture during the manufacturing process. As a result, the oxidation process of copper is greatly decelerated, allowing the sensor to provide more accurate data. Before the test, SubMiniature version A (SMA)-type microwave connectors were soldered to both ends of the ML. A vector network analyzer (VNA, Agilent E8362B) was then connected to the SMA connectors on both ends of the ML for measurement.

The experimental setup for measuring the displacement of the MW is shown in Fig. 5(c). To minimize the impact of the auxiliary fixation device on the experiment, we first determined the position of the MW and then secured it using tape. This ensured that the MW remained in place during testing and prevented displacement caused by external factors, which could otherwise affect the experimental results. Data collection was conducted using the VNA. The VNA emitted precise radio frequency signals and received the reflected and transmitted signals. By performing multiple measurements, the system recorded the phase and amplitude variations between the input and output signals, allowing for the calculation of the corresponding transmission coefficient. From these analytical results, the MW displacement could be obtained. By combining this data with the initial conditions of the MW, sudden displacement changes could also be detected.

3.1. Detection of rotation angle

For the rotation operation, we rotate MW around the center of the DRR. The positive direction of the MW in the y direction is 0° and then rotates counterclockwise. Fig. 6(a) shows a schematic of the sensor used for measuring the RA of the MW. The RA of the MW is simulated in 30°

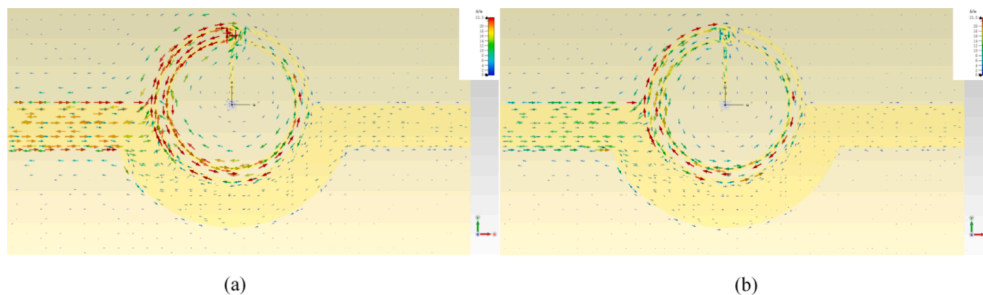


Fig. 4. Simulation of sensor surface current (a) surface current at 0.50 GHz, (b) surface current at 1.18 GHz.

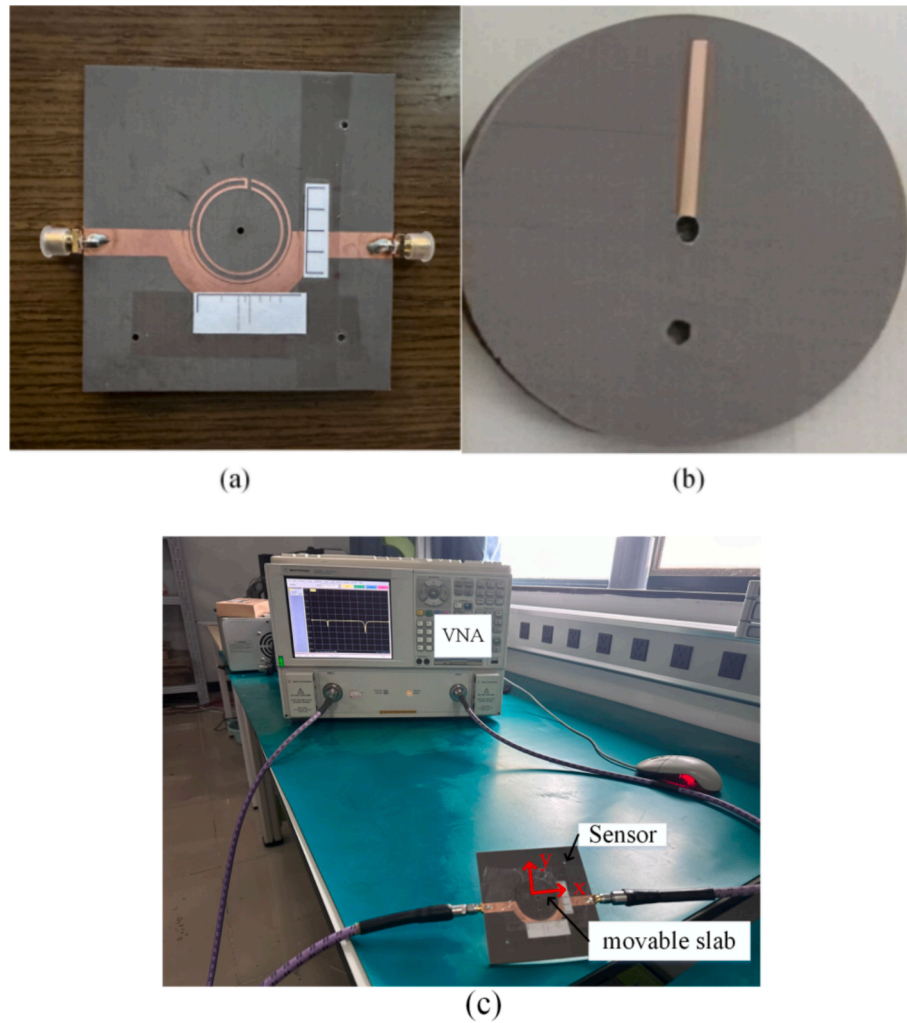


Fig. 5. Picture of the fabricated model (a) fabricated sensor, (b) fabricated movable slab composed of metal wire, (c) experimental test setup.

increments, within the range of 0° - 180° . The simulation and experimental results are shown in Fig. 6(b) and 6(c), where the dashed line represents the simulation curve, and the solid line represents the measurement curve. This convention is used in all the displacement measurement results. The simulation results show that the resonant frequencies corresponding to the MW's rotation from 0° to 180° (in 30° increments) are 0.50, 0.55, 0.62, 0.77, 0.85, 0.98, and 1.22 GHz. This indicates that the resonant frequency gradually shifts to higher values as the RA of the MW increases. The experimental measurement results, shown in Fig. 6(c), indicate the following resonant frequencies from 0° to 180° : 0.64, 0.72, 0.80, 0.90, 1.01, 1.15, and 1.30 GHz. These two figures verify that the simulation and measurement results are consistent, though the measured frequencies are slightly higher than the simulation ones. This discrepancy is primarily attributed to the variation in the intrinsic electrical parameters of the dielectric slab itself. Additionally, slight deviations during the actual model fabrication process and changes in the measurement environment can contribute to this variation.

Sensitivity is typically used to measure the performance of a sensor. It represents the degree of change in the sensor's response caused by a unit change in the measured quantity. The displacement sensitivity of the sensor is defined as follows:

$$S = \frac{\Delta f}{\Delta a} \quad (1)$$

where Δa represents the change in the sensing quantity, which refers to

the change in RA, HD, and VD. Δf represents the variation in the resonant frequency.

Fig. 6(d) shows the simulated resonant frequencies corresponding to different RAs. According to Eq. (1), the slope of the fitted curve represents the angular displacement sensitivity. The calculated sensitivity is 4.06 MHz/degree. The resonant frequency corresponding to each angle can be calculated using the following equation $f = 0.00406\theta + 0.4275$. In addition, the goodness of fit (R^2) is used to evaluate how well the linear regression model fits the simulated data. The goodness of fit can be calculated as follows[29]:

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2 \quad (2)$$

$$SSR = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (3)$$

$$R^2 = 1 - \frac{SSR}{SST} \quad (4)$$

where y_i is the value obtained from each simulation, $\bar{y}$ is the average of all y_i , and $\hat{y}_i$ is the value predicted by the regression model. Therefore, the calculated R^2 value for the regression model is 0.95498. Notably, the closer the value of R^2 is to 1, the better the fit, indicating a stronger linear relationship between the simulated data and the regression model.

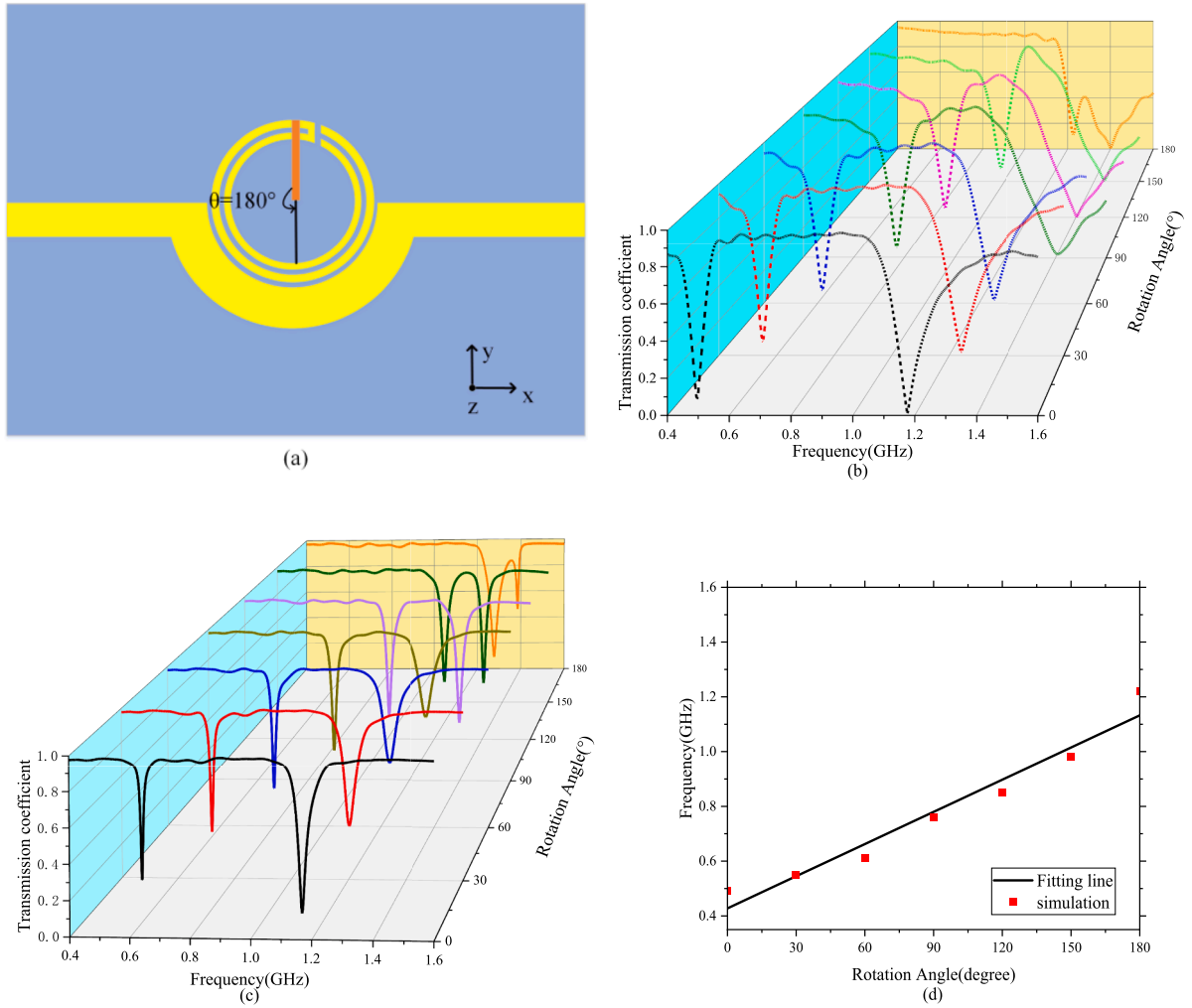


Fig. 6. RA detection (a) schematic diagram of the sensor for measuring RA, (b) simulated variation in the transmission coefficient with the RA, (c) experimental variation in the transmission coefficient with the RA, (d) simulated resonant frequencies corresponding to different RAs.

To further explore the working mechanism of the sensor, the surface current distributions at resonance points were examined for different RAs. Fig. 7(a) shows the surface current distribution at 0.50 GHz when the RA of MW is 0°. It can be seen that the strong current is mainly concentrated on the DRR. Fig. 7(b) displays the surface current distribution at 0.77 GHz when the RA is 90°. At this point, the current in the metal segments of the DRR, which are swept by the MW's rotational movement, becomes very small. This indicates that the equivalent inductance of the DRR has decreased. The value of capacitance is influenced by the geometry of the metal structure and the dielectric constant of the dielectric plate. Since the metal structure of the sensor

and the dielectric constant of the medium plate are already established, it can be assumed that the equivalent capacitance of the DRR does not change significantly during the rotation of the MW. The resonant frequency of the sensor is defined as follows:

$$f = \frac{1}{2\pi\sqrt{LC}} \quad (5)$$

where L and C represent the equivalent inductance and equivalent capacitance of the DRR, respectively. From Eq. (5), it can be concluded that as L decreases, the resonant frequency increases. Therefore, as the RA increases, the resonant frequency shifts to higher values.

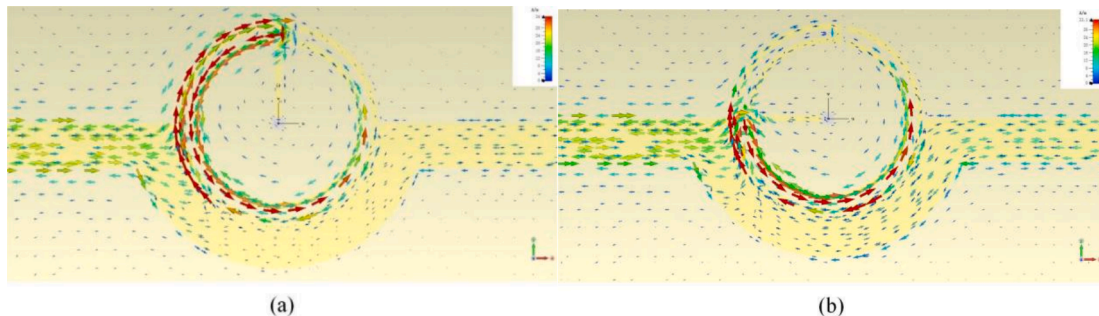


Fig. 7. Simulation of sensor surface current (a) surface current at 0°, (b) surface current at 90°.

3.2. Detection of vertical displacement

For detecting VD, we keep the center of the movable slab aligned with the center of the DRR along the y-axis. The MW points in the negative direction of x and moves from top to bottom. Fig. 8(a) illustrates a schematic of the sensor used to measure the VD of MW. Taking 6 mm as the step size, the VD of the MW is simulated from 0 mm to 18 mm. The simulation results are presented in Fig. 8(b). The resonant frequencies corresponding to the movement of MW from 0 mm to 18 mm (with 6 mm increments) are observed to be 0.67, 0.77, 0.85, and 0.91 GHz. The measurement results are shown in Fig. 8(c), where the resonant frequencies corresponding to the same displacements are 0.73, 0.82, 0.88, and 1.04 GHz, respectively. A comparison between Fig. 8(b) and 8(c) reveals that the simulation results are in excellent agreement with the measurement results. As the MW's VD increases, the resonant frequency gradually shifts towards higher frequencies.

Fig. 8(d) shows the simulated resonant frequencies associated with different VDs. According to Eq. (1), the slope of the fitted curve represents a sensitivity of 18.50 MHz/mm. The corresponding fitting equation is $f = 0.0185y + 0.561$. The goodness of fit (R^2) for the fitted curve is 0.98607.

Fig. 9 shows the simulated surface current distribution when the

sensor is used to measure the VD. Fig. 9(a) shows the surface current distribution at 0.77 GHz when the MW is positioned at $y = 6$ mm. The strong current is primarily concentrated on the DRR. Fig. 9(b) depicts the surface current distribution at 0.85 GHz when the MW is positioned at $y = 12$ mm. In this case, the current in the section of the DRR scanned by the MW decreases, leading to a reduction in the L of the DRR. However, the C of the DRR remains unchanged. According to Eq. (5), the resonant frequency increases as L decreases. Thus, an increase in VD causes a shift in the resonant frequency toward higher values.

3.3. Detection of horizontal displacement

For the detection of HD, we keep the center of the movable slab on the same x-axis as the center of the DRR. The MW points in the y direction and moves from left to right. Fig. 10 shows a schematic of the sensor used to measure the HD of the MW. The HD test was conducted in two stages.

In the first stage, the displacement was simulated with a step size of 7 mm, ranging from 0 mm to 14 mm. The simulation results are presented in Fig. 11(a). The resonant frequencies corresponding to the MW's HD of 0, 7, and 14 mm are observed to be 0.71, 0.55, and 0.48 GHz. Fig. 11(b) shows that the measured resonant frequencies

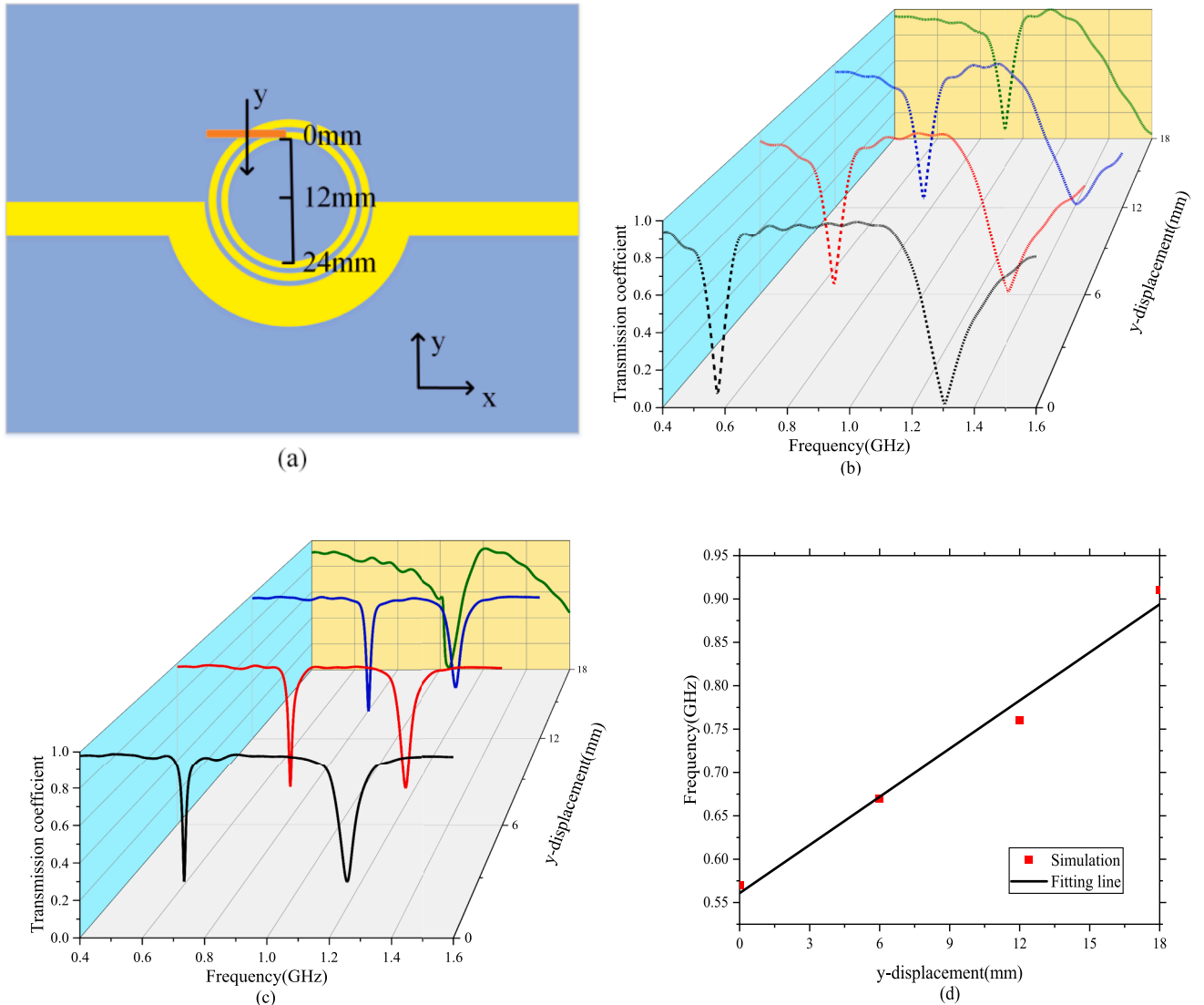


Fig. 8. VD detection a) schematic diagram of the sensor for measuring VD, (b) simulated variation in the transmission coefficient with the VD, (c) experimental variation in the transmission coefficient with the VD, (d) simulated resonant frequencies corresponding to different VDs.

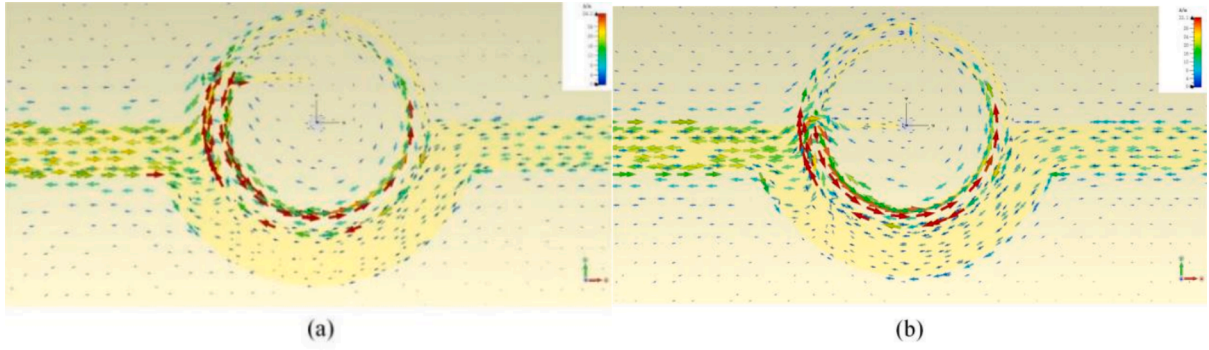


Fig. 9. Simulation of sensor surface current (a) surface current at $y = 6$ mm, (b) surface current at $y = 12$ mm.

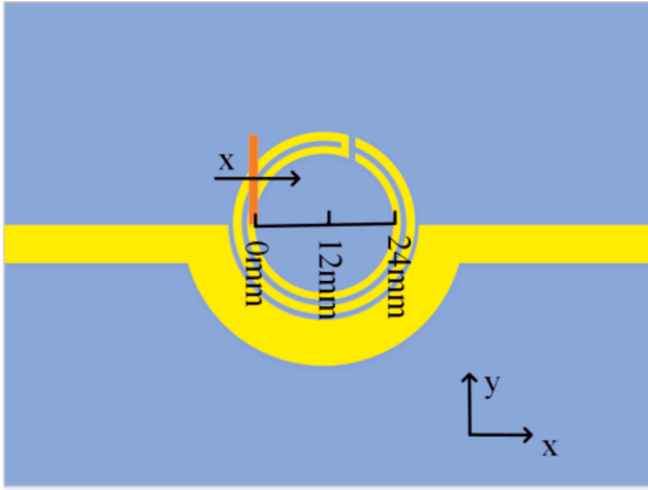


Fig. 10. Horizontal displacement test.

corresponding to the same displacements are 0.87, 0.72, and 0.64 GHz, respectively. In the second stage, the HD of the MW was changed from 15 mm to 24 mm with a step size of 3 mm. The simulation results are presented in Fig. 11(c). The simulated resonant frequencies corresponding to the MW's HD of 15, 18, 21, and 24 mm are 0.88, 0.96, 1.03, and 1.15 GHz, respectively. Fig. 11(d) shows that the measured resonant frequencies corresponding to the same HDs are 1.16, 1.20, 1.25, and 1.28 GHz, respectively. A comparison of the entire simulated and measured HD curves reveals the following trends. Within the range of 0 mm to 14 mm, the resonant frequency shifts to lower values as the HD increases. An opposite trend is observed within the range of 15 mm to 24 mm, where the resonant frequency gradually shifts toward higher values with the increase in HD. Further, the simulation and measurement results are in good agreement with each other.

Fig. 12(a) and 12(b) depict the simulated variations in the resonant frequency with the HD in the range of 0–14 mm and 15–24 mm, respectively. According to Eq. (1), the sensitivity of the sensor within the range of 0–14 mm is 16.4 MHz/mm. The fitting formula for this range is $f = -0.0164x + 0.695$, and the R^2 is 0.95144. The sensitivity within the range of 15–24 mm is 29.6 MHz/mm. The corresponding fitting equation is $f = 0.0296x + 0.874$, and the R^2 value is 0.99323, indicating excellent fitting.

According to the simulation and measurement results for HD, the variation trends of simulated surface current within the range of 0–14 mm are similar to those observed in the RA and VD measurement. Therefore, no additional analysis is provided here for the 0–14 mm range. Fig. 13 illustrates the surface current distribution at two positions for HD detection. Specifically, Fig. 13(a) shows the surface current distribution at 0.88 GHz when the MW is positioned at 15 mm, while

Fig. 13(b) shows the surface current distribution at 1.03 GHz when the MW is positioned at 21 mm. It can be observed that as the MW shifts from 15 mm to 21 mm, the current passing through the DRR decreases. Despite the movement, strong current is still concentrated on the DRR, but the current in the metal segments of the DRR swept by the MW decreases significantly. This reduction in current indicates a decrease in the L of the DRR. As mentioned previously, the C of the DRR is considered to remain relatively unchanged during this process. According to Eq. (5), it is concluded that as L decreases, the resonant frequency increases. Therefore, the resonant frequency shifts to higher values as the HD increases within the range of 15 to 21 mm.

3.4. Influence of structural parameters on the sensor

The influence of the DRR width (k) on the sensor performance is studied using the RA as an example. The value of k is increased from 0.3 mm to 1.2 mm in 0.3 mm increments, and the corresponding transmission curves are simulated, which are shown in Fig. 14(a). It is evident that as k increases, the resonant frequencies corresponding to the two resonant peaks shift toward higher values. Fig. 14(b) shows the transmission curves for different k values after the MW is rotated by 180°. The calculated sensitivities corresponding to different k values are as follows: 3.96 MHz/degree for $k = 0.3$ mm, 3.98 MHz/degree for $k = 0.6$ mm, 4.06 MHz/degree for $k = 0.9$ mm, and 4.04 MHz/degree for $k = 1.2$ mm. Based on these results, a ring width of $k = 0.9$ mm is chosen for optimal sensor performance.

Next, the effect of the gap (c) between the DRR and the ML on the sensor performance is explored. The value of c is increased from 0.2 mm to 0.6 mm in steps of 0.2 mm, and the simulated transmission curves are shown in Fig. 15. The results show that although the quality factor (Q) values of the resonance dips at c of 0.4 and 0.6 mm are higher, the energy loss at the low-frequency dip is higher. By contrast, for $c = 0.2$ mm, the resonance dips are deeper, indicating that the energy transfer between the electromagnetic waves and the metamaterial structure is more efficient at this gap. Therefore, based on these observations, a gap of $c = 0.2$ mm is chosen as the optimal value.

In practical applications, the width of the measured object can vary. Therefore, to investigate the impact of the MW width on the sensor performance, simulations were performed with a step size of 0.5 mm, varying the MW width (s) from 0.5 mm to 2.0 mm. The resulting transmission curves are shown in Fig. 16. It can be seen that as s increases, the low-frequency dip in the transmission curve remains unchanged, while the high-frequency dip gradually shifts to lower frequencies. Notably, the sensor detects displacement based on the frequency shift of the low-frequency dip, and the corresponding resonant frequency remains unaffected by the MW width. Therefore, the sensor's performance is not influenced by changes in the width of the MW, making it suitable for practical applications where object dimensions may vary.

To demonstrate the superiority and versatility of the proposed

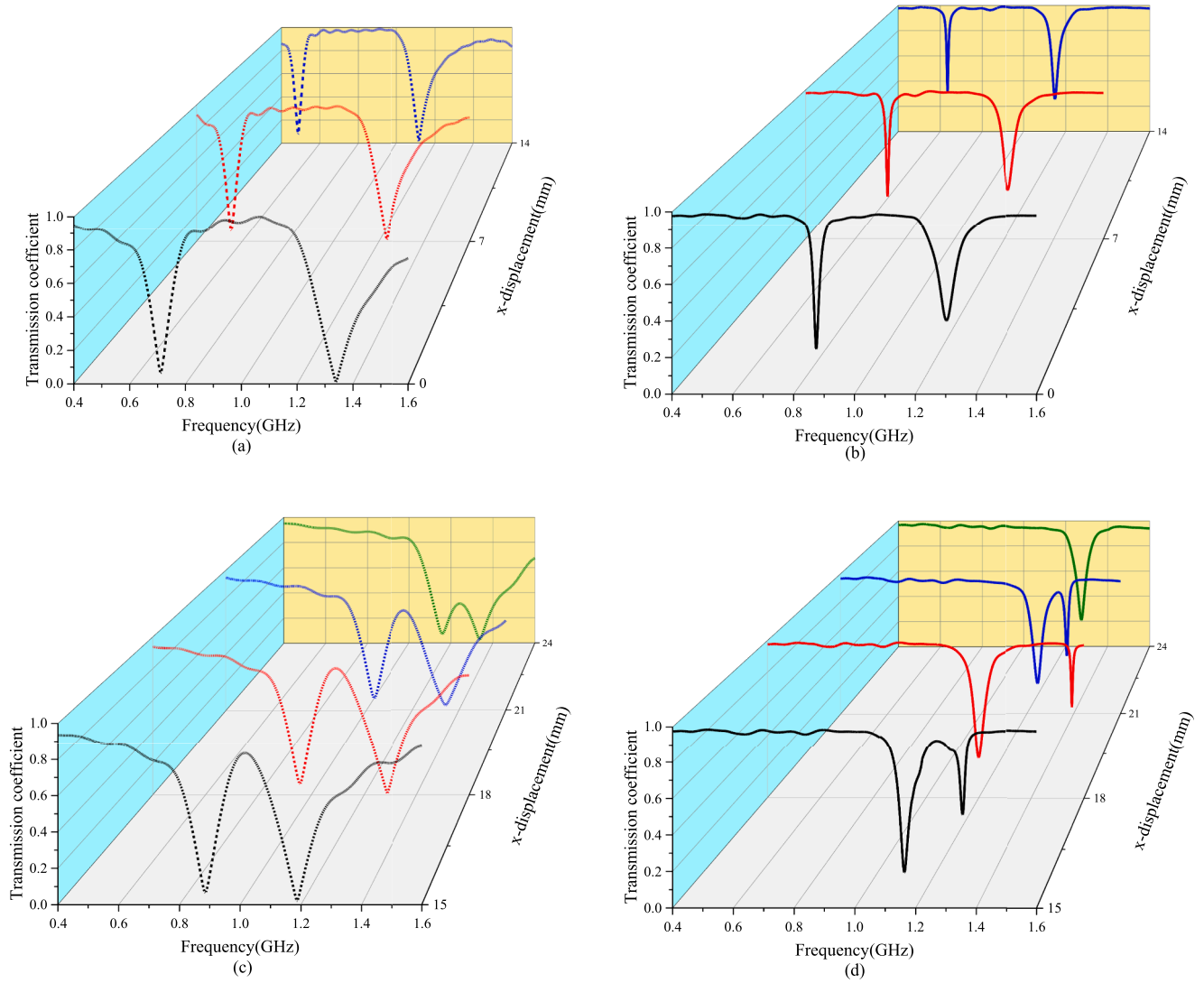


Fig. 11. HD detection (a) simulated variation in the transmission coefficient with the HD (0–14 mm), (b) measured variation in the transmission coefficient with the HD (0–14 mm), (c) simulation variation in the transmission coefficient with the HD (15–24 mm), (d) measured variation in the transmission coefficient with the HD (15–24 mm).

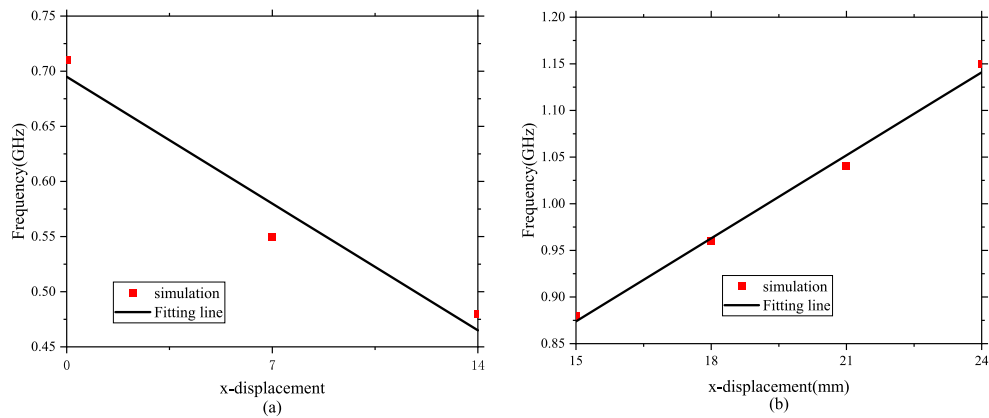


Fig. 12. HD detection (a) simulated resonant frequencies corresponding to different HDs in the range of 0–14 mm, (b) simulated resonant frequencies corresponding to different HDs in the range of 15–24 mm.

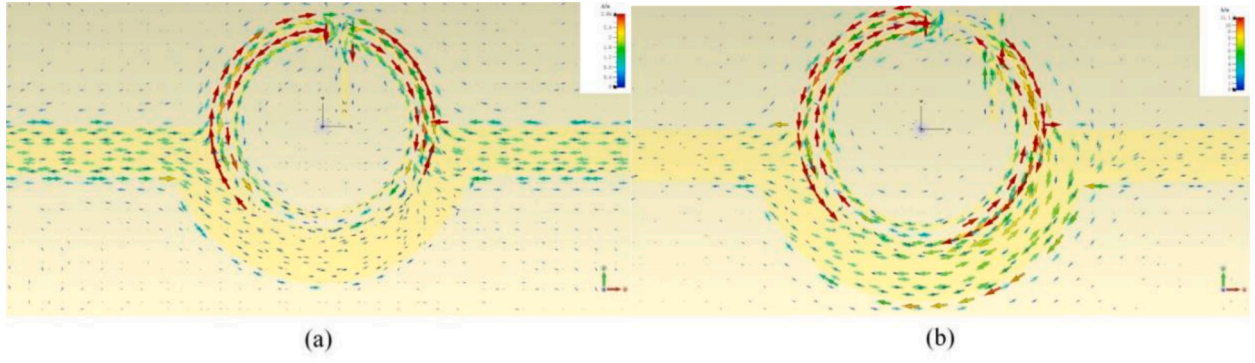


Fig. 13. Simulation of sensor surface current (a) surface current at $x = 15$ mm position, (b) surface current at $x = 21$ mm position.

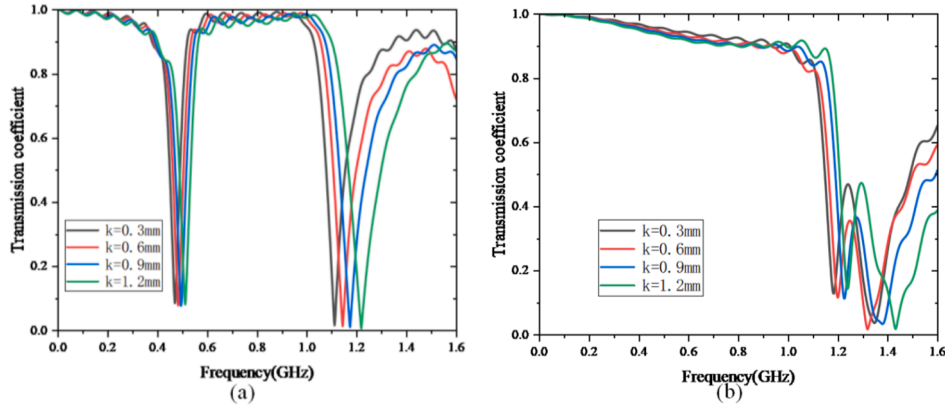


Fig. 14. Effect of DRR ring width on the sensor performance (a) 0° position, (b) 180° position.

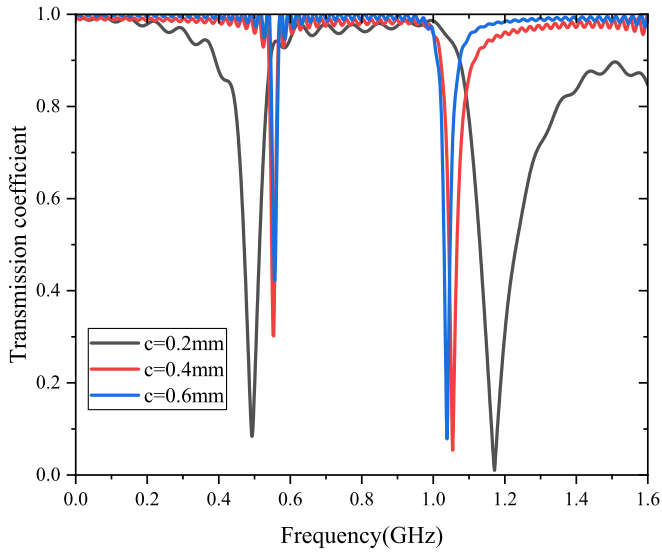


Fig. 15. Effect of the gap between DRR and ML on the sensor performance.

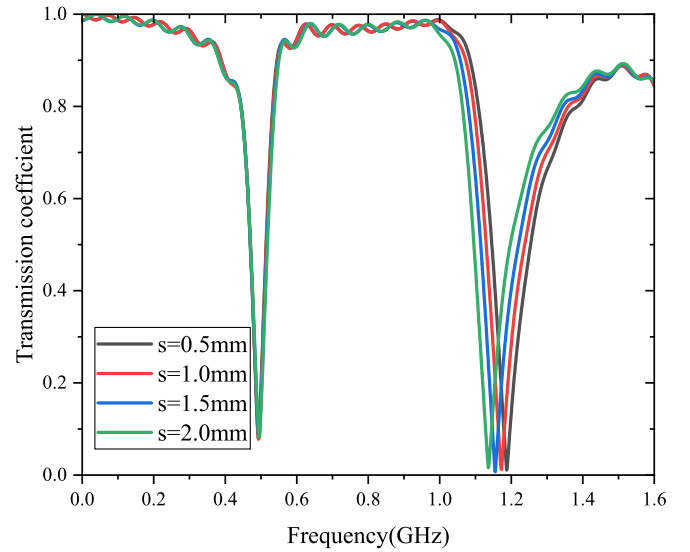


Fig. 16. Effect of MW width on the sensor performance.

sensor, a comparison is made with previously reported displacement sensors, as shown in Table 1. The sensors in references [8,9], and [10] are designed for RA measurement only. Specifically, the sensors in references [9] and [10] are capable of detecting a RA of up to 180° . The sensor in reference [12] is capable of detecting both RA and HD. By contrast, the proposed sensor facilitates multi-parameter detection. Specifically, it can measure three parameters: RA, HD, and VD. Furthermore, it demonstrates higher sensitivity in detecting RA

compared to similar types of sensors, enhancing its effectiveness in practical SHM applications.

4. Conclusion

In this study, an effective metamaterial sensor based on an ML-loaded DRR for multi-parameter measurement was proposed. The working mechanism of the sensor was analyzed by establishing an

Table 1
Comparison of displacement sensors.

Ref	Type of sensor	Detectable parameter	Sensitivity
[8]	CSRR with TLY-5	Rotation angle(90°)	2.37 MHz/degree
[9]	Microstrip: U-shaped	Rotation angle(180°)	1.81 MHz/degree
[10]	Microstrip: TSI	Rotation angle(180°)	1.24 MHz/degree
[11]	SRR	vertical displacement	30.1 MHz/mm(0–9 mm)
[12]	Microstrip: SRR	vertical displacementHorizontal displacement	18.75 dB/mm(0.1–0.8 mm) 18.75 dB/mm(0.1–0.8 mm)
This work	Microstrip: circular	Rotation angle (180°)vertical displacementHorizontal displacement	4.06 MHz/degree 18.9 MHz/mm 16.4 MHz/mm(0–14 mm) 29.6 MHz/mm(15–24 mm)

equivalent circuit model. The feasibility and effectiveness of the sensor model were validated through a combination of simulations and experimental measurements. The displacement operation of the MW in the sensor structure altered the alignment between the MW and the DRR, which modified the equivalent inductance of the DRR. This change in inductance led to a corresponding shift in the resonant frequency, facilitating the detection of three parameters with high sensitivity: RA (0°–180°), VD (0–18 mm), and HD (0–14 mm, 15–18 mm). The multi-parameter sensing capability makes the proposed sensor particularly useful in SHM applications, such as incline detection in steel bridges and longitudinal and transverse displacement monitoring in metal pipelines.

CRedit authorship contribution statement

Zhaoyang Shen: Conceptualization. **Fan Luo:** Formal analysis. **Qin Qin:** Software. **Linyan Guo:** Validation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.measurement.2025.116652>.

Data availability

No data was used for the research described in the article.

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